HW6 何衍宁 HE Yanning 12311512

1. A Si p+-n-p transistor has impurity concentrations of 5×10^{18}, 2×10^{17} and 10^{16} cm^{-3} in the emitter, base, and collector, respectively. The base width W_B is 1 μm and the device cross-sectional area is 2 mm. When the emitter-base junction is forward-biased to 0.5 V and the base-~~collector~~ junction is reverse-biased to 5 V, calculate (a) the neutral base width and (b) the minority carrier concentration at x=0 in the neutral base.

(Ans. 0.904 μm and 1.13 ×10^{11} cm^{3})

$N_{A,E} = 5\times10^{18}\,cm^{-3}$, $N_{D,B} = 2\times10^{17}\,cm^{-3}$

$N_{A,C} = 10^{16}\,cm^{-3}$, $W_B = 1\,\mu m$,

$V_{EB} = +0.5V$, $V_{BC} = -5V$, $kT = 0.026\,eV$

a) $$W = \sqrt{\frac{2\varepsilon_s}{q}\cdot\left(\frac{1}{N_A}+\frac{1}{N_D}\right)\cdot(V_{bi}-V_{app})}$$

$$V_{bi} = \frac{kT}{q}\cdot\ln\left(\frac{N_A\cdot N_D}{n_i^2}\right) \quad \leftarrow n_i = 1.5\times10^{10}\,cm^{-3}$$

$$\Rightarrow V_{bi,EB} = 0.026\times\ln\left(\frac{5\times10^{18}\times2\times10^{17}}{(1.5)^2\times10^{20}}\right)$$

$$= 0.937V$$

$$\Rightarrow V_{bi,BC} = 0.026\cdot\ln\left(\frac{2\times10^{17}\times10^{16}}{(1.5\times10^{10})^2}\right)$$

$$= 0.775V$$

$\Rightarrow$ EB : $V_{EB} = +0.5V$ $\qquad \varepsilon_0 = 8.85\times10^{-14}\,F/cm$

$$W_{EB,tot} = \sqrt{\frac{2\cdot11.7\varepsilon_0}{1.6\times10^{-19}}\left(\frac{1}{5\times10^{18}}+\frac{1}{2\times10^{17}}\right)(0.937-0.5)}$$

$$= 5.42\times10^{-6}\,cm$$

$$W_{EB,n} = W_{EB,tot}\cdot\frac{N_{A,E}}{N_{A,E}+N_{D,B}} \approx 5.21\times10^{-6}\,cm$$

$\Rightarrow BC \quad V_r = -5V$

$$W_{BC,tot} = \sqrt{\frac{2 \cdot \varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_{bi,BC} + V_r)}$$

$$= 8.856 \times 10^{-5}\ cm$$

$$W_{BC,n} = W_{BC,tot} \cdot \frac{N_{A,C}}{N_{A,C} + N_{D,B}} \simeq 4.22 \times 10^{-6}\ cm$$

$\Rightarrow W_{B,neutral} = W_B - W_{BC,n} - W_{EB,n} \simeq 0.904\ \mu m$

b) $$P_n(0) = P_{no} \cdot \exp\left(\frac{V_{EB}}{kT}\right)$$

$$P_{no} = \frac{n_i^2}{N_{D,B}} \Rightarrow P_n(0) = \frac{n_i^2}{N_{D,B}} \cdot e^{\frac{0.5}{0.026}} \simeq 1.13 \times 10^{11}\ cm^{-3}$$

2. For the transistor in Problem 1, the diffusion constants of minority carriers in the emitter, base, and collector are 52, 40, and 115 cm^2/s, respectively; and the corresponding lifetimes are 10^{-8}, 10^{-7}, and 10^{-6} s. Find the current component I_{Ep}, I_{Cp}, I_{En}, I_{Cn}, and I_{BB}.

(Ans. 1.596×10^{-5} A, 1.596×10^{-5} A, 1.04×10^{-7} A, 3.196×10^{-14} A, 0)

$D_{n,E} = 52\ cm^2/s, \quad \tau_{n,E} = 10^{-8}\ s$

$D_{p,B} = 40\ cm^2/s, \quad \tau_{p,B} = 10^{-7}\ s$

$D_{n,C} = 115\ cm^2/s, \quad \tau_{n,C} = 10^{-6}\ s$

$$\Rightarrow L_E = \sqrt{D_{n,E} \cdot \tau_{n,E}} \simeq 7.21 \times 10^{-4}\ cm$$

$$L_B = \sqrt{D_{p,B} \cdot \tau_{p,B}} \simeq 2.0 \times 10^{-3}\ cm$$

$$L_C = \sqrt{D_{n,C} \cdot \tau_{n,C}} \simeq 1.07 \times 10^{-2}\ cm$$

$\Rightarrow A = 2\ mm^2 = 0.02\ cm^2, \quad P_n(0) = 1.13 \times 10^{11}\ cm^{-3}$

$$\Rightarrow \quad I_{Ep} = q \cdot A \cdot D_p \frac{p_n(0) - p_n(W_{Bn})}{W_{Bn}}$$

since BC junction is rb, $p_n(W_{Bn}) \simeq 0$

$$I_{Ep} = (1.6 \times 10^{-19})(0.02)(40) \cdot \frac{1.13 \times 10^{11}}{9.04 \times 10^{-5}}$$

$$= 1.596 \times 10^{-5} \text{ A}$$

$$\Rightarrow \quad I_{Cp} \simeq I_{Ep} = 1.596 \times 10^{-5} \text{ A}$$

$$\Rightarrow \quad n_{p0,E} = \frac{n_i^2}{N_{AE}} = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{18}} = 4.5 \times 10^{1} \text{ cm}^{-3}$$

$$n_p(0) = n_{p0,E} \cdot \exp\left(\frac{qV_{EB}}{kT}\right) = (4.5 \times 10^{1}) \cdot \exp\left(\frac{0.5}{0.026}\right) = 1.01 \times 10^{10} \text{ cm}^{-3}$$

$$I_{En} = q \cdot A \cdot D_{nE} \cdot \frac{n_p(0)}{L_E}$$

$$= (1.6 \times 10^{-19}) \cdot (0.02)(52) \frac{1.01 \times 10^{10}}{7.21 \times 10^{-4}}$$

$$= 1.04 \times 10^{-7} \text{ A}$$

$$\Rightarrow \quad I_{Cn} = qA \cdot \frac{n_i^2}{N_{AC}} \cdot \frac{D_{nC}}{L_C} \simeq 3.196 \times 10^{-14} \text{ A}$$

$$\Rightarrow \quad I_{BB} = 0$$

3. Using the results obtained from problem 1 and 2, (a) find the terminal currents I_E, I_C, and I_B, of the transistor; (b) calculate emitter efficiency, base transport factor, common-base current gain, and common-emitter current gain. (c) Comment on how the emitter efficiency and base transport can be improved.

(Ans.(a) 1.606×10^{-5} A, 1.596×10^{-5} A, 1.041×10^{-7}A, (b) 0.9938, 1 ,0.9938, 160.3)

a) $I_E = I_{Ep} + I_{En} = 1.606 \times 10^{-5}\,A$

$I_C = I_{Cp} + I_{Cn} \simeq 1.596 \times 10^{-5}\,A$

$I_B = I_E - I_C = 1.041 \times 10^{-7}\,A$

b) $\gamma = \dfrac{I_{Ep}}{I_{Ep} + I_{En}} = \dfrac{1.596}{1.596 + 1.04 \times 10^{-2}} \simeq 0.9938$

$\alpha_T = \dfrac{I_{Cp}}{I_{Ep}} = 1$

$\alpha = \dfrac{I_C}{I_E} = \dfrac{1.596}{1.606} \simeq 0.9938 \Rightarrow \alpha = \gamma \alpha_T$

$\beta = \dfrac{\alpha}{1-\alpha} \simeq 160.3$

c) Improve γ / α_T

γ: ① Increasing emitter doping ② Reducing base doping ③ Wide bandgap emitter

α_T: ① Reducing base width ② Increasing base carrier lifetime

4. A Si transistor has D of 10 cm^2/s and W of 0.5 μm. Find the cutoff frequencies for the transistor with a common base current gain of 0.998. Neglect the emitter and collector delays. To design a bipolar transistor with 5 GHz cutoff frequency, what the neutral base width W will be?

(Ans:(a) f= 1.27 GHz and f_β= 2.55 MHz;(b) 0.252 μm)

$D = 10\,cm^2/s$, $W = 0.5\,\mu m$, $\alpha = 0.998$

a) $\tau_B = \dfrac{W^2}{2D} \approx \dfrac{(5\times10^{-5})^2}{20} = 1.25\times10^{-10}\,s$

$f_\alpha = \dfrac{1}{2\pi\tau_B} = \dfrac{1}{2\pi\cdot1.25\times10^{-10}} = 1.27\ GHz$

$f_\beta = (1-\alpha)\cdot f_\alpha = (1-0.998)\cdot(1.27\ GHz) = 2.55\ MHz$

b) In order to $f_\alpha = 5\ GHz$

$f_\alpha = \dfrac{D}{\pi W^2} \rightarrow \quad W = \sqrt{\dfrac{D}{\pi f_\alpha}} = \sqrt{\dfrac{10}{\pi\cdot 5GHz}}$

$\Rightarrow W = 0.252\ \mu m$